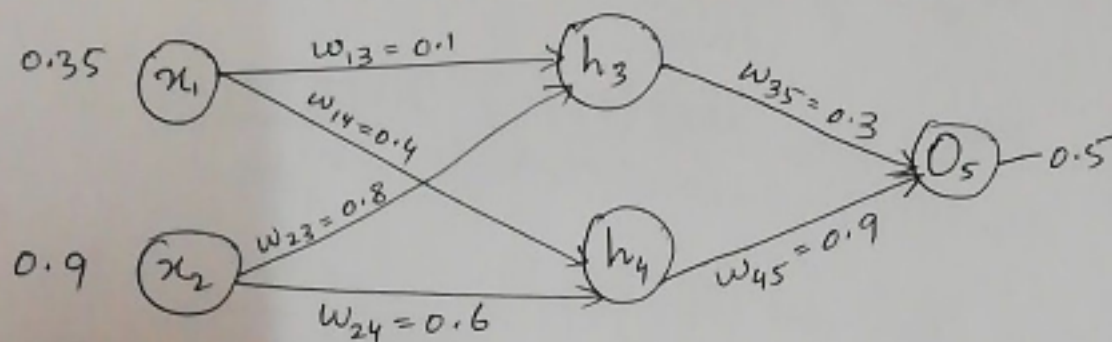


Question # 1: Backpropagation:-



Solution:

$$\text{net}(h_3) = x_1 w_{13} + x_2 w_{23} = (0.35)(0.1) + (0.9)(0.8)$$

$$\text{net}(h_3) = 0.755$$

$$\text{net}(h_4) = x_1 w_{14} + x_2 w_{24} = (0.35)(0.4) + (0.9)(0.6)$$

$$\text{net}(h_4) = 0.68$$

$$\text{out}(h_3) = \frac{1}{1 + e^{-\text{net}(h_3)}} = \frac{1}{1 + e^{-0.755}} = 0.68$$

$$\text{out}(h_4) = \frac{1}{1 + e^{-\text{net}(h_4)}} = \frac{1}{1 + e^{-0.68}} = 0.664$$

$$\begin{aligned} \text{net}(O_5) &= \text{out}(h_3) w_{35} + \text{out}(h_4) w_{45} \\ &= (0.68)(0.3) + (0.664)(0.9) \\ &= 0.8016 \end{aligned}$$

$$\text{out}(O_5) = \frac{1}{1 + e^{-\text{net}(O_5)}} = \frac{1}{1 + e^{-0.8016}}$$

$$\text{out}(O_5) = 0.6903$$

$$\text{Error} = \frac{1}{2} (\text{target} - \text{O/P})^2 = \frac{1}{2} (0.5 - 0.6903)^2$$

$$\boxed{\text{Error} = 0.01811}$$

Now, update weights :-

update w_{35} :

$$w'_{35} = w_{35} - \eta \frac{\partial E_T}{\partial w_{35}}$$

$$\frac{\partial E_T}{\partial w_{35}} = \frac{\partial E_T}{\partial \text{out}(o_5)} \times \frac{\partial \text{out}(o_5)}{\partial \text{net}(o_5)} \times \frac{\partial \text{net}(o_5)}{\partial w_{35}}$$

$$\begin{aligned} \frac{\partial E_T}{\partial \text{out}(o_5)} &= -[o_5 - \text{out}(o_5)] \\ &= -(0.5 - 0.6903) = 0.1903 \end{aligned}$$

$$\begin{aligned} \frac{\partial \text{out}(o_5)}{\partial \text{net}(o_5)} &= \text{out}(o_5) [1 - \text{out}(o_5)] \\ &= (0.6903)(1 - 0.6903) = 0.2138 \end{aligned}$$

$$\frac{\partial \text{net}(o_5)}{\partial w_{35}} = \text{out}(h_3) = 0.68$$

$$\begin{aligned} \frac{\partial E_T}{\partial w_{35}} &= (0.1903)(0.2138)(0.68) \\ &= 0.0276 \end{aligned}$$

$$w'_{35} = 0.3 - (1)(0.0276)$$

$$\boxed{w'_{35} = 0.2723}$$

update w_{45} :

$$w'_{45} = w_{45} - \eta \frac{\partial E_T}{\partial w_{45}}$$

$$\frac{\partial E_T}{\partial w_{45}} = \frac{\partial E_T}{\partial \text{out}(o_5)} \times \frac{\partial \text{out}(o_5)}{\partial \text{net}(o_5)} \times \frac{\partial \text{net}(o_5)}{\partial w_{45}}$$

$$\frac{\partial E_T}{\partial w_{45}} = (0.1903)(0.2138)(0.664) = 0.027$$

$$w'_{45} = 0.9 - 0.027$$

$$w'_{45} = 0.8730$$

update w_{13} :

$$w'_{13} = w_{13} - \eta \frac{\partial E_T}{\partial w_{13}}$$

$$\frac{\partial E_T}{\partial w_{13}} = \frac{\partial E_T}{\partial \text{out}(h_3)} \times \frac{\partial \text{out}(h_3)}{\partial \text{net}(h_3)} \times \frac{\partial \text{net}(h_3)}{\partial w_{13}}$$

$$\frac{\partial E_T}{\partial \text{out}(h_3)} = \frac{\partial E_T}{\partial \text{out}(o_5)} \times \frac{\partial \text{out}(o_5)}{\partial \text{net}(o_5)} \times \frac{\partial \text{net}(o_5)}{\partial \text{out}(h_3)}$$

$$\frac{\partial E_T}{\partial \text{out}(o_5)} = 0.1903$$

$$\frac{\partial \text{out}(o_5)}{\partial \text{net}(o_5)} = 0.2138$$

$$\frac{\partial \text{net}(o_5)}{\partial \text{out}(h_3)} = w_{35} = 0.3$$

$$\frac{\partial E_T}{\partial \text{out}(h_3)} = (0.1903)(0.2138)(0.3) = 0.0122$$

$$\frac{\partial \text{out}(h_3)}{\partial \text{net}(h_3)} = \text{out}(h_3) [1 - \text{out}(h_3)]$$

$$= 0.68 (1 - 0.68) = 0.2176$$

$$\frac{\partial \text{net}(h_3)}{\partial w_{13}} = x_1 = 0.35$$

$$\frac{\partial E_T}{\partial w_{13}} = (0.0122)(0.2176)(0.35)$$

$$= 0.000929152$$

$$W'_{13} = 0.1 - 0.000929152$$

$$\boxed{W'_{13} = 0.099}$$

update W_{14} :

$$W'_{14} = W_{14} - \eta \frac{\partial E_T}{\partial W_{14}}$$

$$\frac{\partial E_T}{\partial W_{14}} = \frac{\partial E_T}{\partial \text{out}(h_4)} \times \frac{\partial \text{out}(h_4)}{\partial \text{net}(h_4)} \times \frac{\partial \text{net}(h_4)}{\partial W_{14}}$$

$$\frac{\partial E_T}{\partial \text{out}(h_4)} = \frac{\partial E_T}{\partial \text{out}(o_5)} \times \frac{\partial \text{out}(o_5)}{\partial \text{net}(o_5)} \times \frac{\partial \text{net}(o_5)}{\partial \text{out}(h_4)}$$

$$\frac{\partial E_T}{\partial \text{out}(h_4)} = (0.1903)(0.2138)(0.9) = 0.0366$$

$$\begin{aligned} \frac{\partial \text{out}(h_4)}{\partial \text{net}(h_4)} &= \text{out}(h_4) (1 - \text{out}(h_4)) \\ &= 0.664 (1 - 0.664) \\ &= 0.223 \end{aligned}$$

$$\frac{\partial \text{net}(h_4)}{\partial W_{14}} = x_1 = 0.35$$

$$\begin{aligned} \frac{\partial E_T}{\partial W_{14}} &= (0.0366)(0.223)(0.35) \\ &= 0.002858 \end{aligned}$$

$$W'_{14} = 0.4 - 0.002858$$

$$\boxed{W'_{14} = 0.397}$$

Update W_{23} :

$$W'_{23} = W_{23} - \eta \frac{\partial E_T}{\partial W_{23}}$$

$$\frac{\partial E_T}{\partial W_{23}} = \frac{\partial E_T}{\partial \text{out}(h_3)} \times \frac{\partial \text{out}(h_3)}{\partial \text{net}(h_3)} \times \frac{\partial \text{net}(h_3)}{\partial W_{23}}$$

$$\frac{\partial E_T}{\partial \text{out}(h_3)} = \frac{\partial E_T}{\partial \text{out}(o_5)} \times \frac{\partial \text{out}(o_5)}{\partial \text{net}(o_5)} \times \frac{\partial \text{net}(o_5)}{\partial \text{out}(h_3)}$$

$$\frac{\partial E_T}{\partial \text{out}(h_3)} = (0.1903)(0.2138)(0.3) = 0.0122$$

$$\frac{\partial \text{out}(h_3)}{\partial \text{net}(h_3)} = 0.2176$$

$$\frac{\partial \text{net}(h_3)}{\partial W_{23}} = x_2 = 0.9$$

$$\frac{\partial E_T}{\partial W_{23}} = (0.0122)(0.2176)(0.9) = 0.0024$$

$$W'_{23} = 0.8 - 0.0024$$

$$W'_{23} = 0.79761$$

Update W_{24} :

$$W'_{24} = W_{24} - \eta \frac{\partial E_T}{\partial W_{24}}$$

$$\frac{\partial E_T}{\partial W_{24}} = \frac{\partial E_T}{\partial \text{out}(h_4)} \times \frac{\partial \text{out}(h_4)}{\partial \text{net}(h_4)} \times \frac{\partial \text{net}(h_4)}{\partial W_{24}}$$

$$\frac{\partial E_T}{\partial W_{24}} = (0.0366)(0.223)(0.9) = 0.00734$$

$$w'_{24} = 0.6 - 0.00734$$

$$w'_{24} = 0.5926$$

Now, all weights are updated.

Then another forward pass is:

$$\text{net}(h_3) = x_1 w'_{13} + x_2 w'_{23}$$

$$= (0.35)(0.099) + (0.9)(0.79761)$$

$$= 0.7525$$

$$\text{out}(h_3) = \frac{1}{1 + e^{-\text{net}(h_3)}} = \frac{1}{1 + e^{-0.7525}} = 0.6797$$

$$\text{net}(h_4) = x_1 w'_{14} + x_2 w'_{24}$$

$$= (0.35)(0.397) + (0.9)(0.5926)$$

$$= 0.6723$$

$$\text{out}(h_4) = \frac{1}{1 + e^{-\text{net}(h_4)}} = \frac{1}{1 + e^{-0.6723}} = 0.662$$

$$\text{net}(o_5) = \text{out}(h_3) w'_{35} + \text{out}(h_4) w'_{45}$$

$$= (0.6797)(0.2723) + (0.662)(0.873)$$

$$= 0.763$$

$$\text{out}(o_5) = \frac{1}{1 + e^{-\text{net}(o_5)}} = \frac{1}{1 + e^{-0.763}} = 0.68201$$

$$\text{Error} = \frac{1}{2} (\text{target} - o/p)^2$$

$$= \frac{1}{2} (0.5 - 0.68201)^2$$

$$\text{Error} = 0.01656$$

Ans